

PHY 412: Statistical Physics

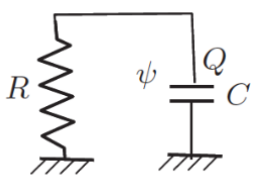
25-26 Semester II

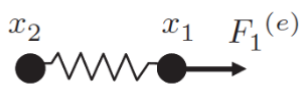
Assignment – 3

Problems in black are for submission; Blue ones are for practice. Due date: March 30th in Class

1. Random walks with a constant drift term will have a net correlation between steps. However, it can be reduced to random walks without drift by shifting to the moving reference frame. Suppose we have a random walk with steps independently drawn from a uniform density $\rho(l)$ on $[0,1]$, but with a nonzero mean $\langle l \rangle = \bar{l} \neq 0$.
 - (a) Argue that the sums $S'_N = \sum_{n=1}^N (l_n - \bar{l})$ describe random walks with zero mean in a moving reference frame.
 - (b) Argue that the variance of these random walks (the squared standard deviation) is the same as the variance $\langle (S_N - \bar{S}_N)^2 \rangle$ of the original random walks.
2. Consider a colloidal particle of radius $0.1 \mu\text{m}$, density 2 g cm^{-3} moving in water (density 1 g cm^{-3} , viscosity 1 mPa.s). Answer the following questions:
 - (a) Calculate the sedimentation velocity V_s of the particle.
 - (b) Calculate the diffusion constant D of the particle.
 - (c) Calculate the average distance $V_s t$ that the particle moves in time t , and its root-MSD $\sqrt{Dt}$ for $t = 1 \text{ ms}, 1 \text{ s}, 1 \text{ hour}, 1 \text{ day}$. Discuss when do we need to consider Brownian motion.
3. Consider the electrical circuit given at the right; the circuit is made of a resistor (resistance R) and a capacitor (capacitance C). Charge Q stored in the capacitor is 0 on average, but can be positive or negative due to the thermal fluctuations of charge carriers. Answer the following questions:

- (a) Write down the equation describing the fluctuating charge $Q(t)$, under the influence of a randomly fluctuating voltage $\psi_r(t)$. Comparing the equation with the Langevin equation, show that the fluctuating voltage satisfies the fluctuation-dissipation relation (also called as Nyquist theorem in electrical engineering context), $\langle \psi_r(t) \psi_r(0) \rangle = 2Rk_B T \delta(t)$.
 - (b) Calculate the time correlation functions $\langle Q(t)Q(0) \rangle$ and $\langle I(t)I(0) \rangle$, where $I(t) = \dot{Q}(t)$ is the current flowing in the resistor.


4. The Langevin equation for a Brownian particle in a harmonic potential $\frac{dx}{dt} = -\frac{k}{\gamma} x + v_r(t) = -\frac{x}{\tau} + v_r(t)$, gives a steady-state solution for $x(t)$ as $x(t) = \int_0^t dt_1 e^{-(t-t_1)/\tau} v_r(t_1)$. Answer the following questions:
 - (a) Calculate $\langle x(t)^2 \rangle$ from the solution $x(t)$ using the time-correlation of random-velocity ($v_r(t) = \frac{F_r(t)}{\gamma}$, $F_r(t)$ is the random force) $\langle v_r(t) v_r(t') \rangle = 2D\delta(t - t')$.
 - (b) Calculate the time correlation function $\langle x(t)x(t') \rangle$ for $t > t'$, and $t < t'$.
 - (c) Show that $\langle \dot{x}(t)\dot{x}(0) \rangle = 2D\delta(t) - \frac{D}{\tau} e^{-|t|/\tau}$ starting from the Langevin equation given above.
5. Consider two Brownian particles connected by a harmonic spring, as shown in the figure at right. The Langevin equations for their positions x_1 and x_2 are written as: $\zeta \frac{dx_1}{dt} = -k(x_1 - x_2) + F_{r1}(t)$, and $\zeta \frac{dx_2}{dt} = -k(x_2 - x_1) + F_{r2}(t)$. Answer the following questions:



- (a) Suppose that a constant force $F_1^{(e)}$ is applied to the particle 1 for $t > 0$. Calculate the average displacements of the particles $\langle x_1(t) - x_1(0) \rangle_{F_1^{(e)}}$ and $\langle x_2(t) - x_2(0) \rangle_{F_1^{(e)}}$ in this situation.
- (b) Discuss the behavior of $\langle (x_1(t) - x_1(0))^2 \rangle$ at short time ($t \ll \zeta/k$) and at long time ($t \gg \zeta/k$).
6. Consider a Brownian particle in one dimension along x -axis. There is a source of propulsion that drives the particle at a constant speed along the positive x direction. Hence the velocity of the particle at any time t can be given by, $v(t) = v_r(t) + v_x$, where $v_r(t)$ is the random velocity associated with the random force that the particle experiences, and v_x is the time-independent propulsion speed. For this system
- Calculate and comment on the velocity autocorrelation function $\langle v(t)v(0) \rangle$
 - From the velocity autocorrelation, calculate the force autocorrelation $\langle \xi(t)\xi(0) \rangle$
 - Comment on the validity of fluctuation-dissipation theorem (FDT) from (ii)
 - Calculate the displacement autocorrelation $\langle (x(t_1) - x(0))(x(t_2) - x(0)) \rangle$
 - From displacement autocorrelation calculate and comment on MSD $\langle (x(t_1) - x(0))^2 \rangle$